SAFAL – Software with integrated AI for Adaptive Learning

Cline Hackathon Round 1 technical documentation by ***Team Skill Issue***

Our proposal

We want to create a school that provides AI-based education. All students will follow the same curriculum, but they will study it individually and at their own pace. The system will identify each student's learning gaps and try to close them using different explanations, micro-tests, and customized home assignments. Each student will be given their own tab, which will be restricted to running only the software product we create. The tabs will be kept in the school, and students can access their AI tutors at home by using their credentials to log in.

Every topic will first be taught through a recorded lecture, followed by a reading assignment. This will be the same for all students. After completing the reading assignment, the AI tutor will ask the student if they have any doubts and help clear them. Once the student feels that they have no more doubts, they will take a micro-test based on the topic. Their performance in the test will help the system understand how well they have learned the topic. The system will then analyze the students' answers and try to understand why they made mistakes. Based on the student's learning and mistake patterns, the AI tutor will create a customized home assignment for each student. These assignments will focus on fixing their weaknesses and misconceptions.

Every student has a different mind and different learning abilities. Students can also have different levels of aptitude in different subjects. Some students may understand certain topics faster than others. Because of this, students in this school will not necessarily move through the curriculum at the same speed. The curriculum will be divided into terms, with each term having a fixed portion of the curriculum. At the end of each term, all students will take a similar exam paper. This will allow their performance to be measured using the same standard. If a student can complete their term's syllabus before the end of the term, they can be given a vacation. During this time, they can explore their interests and can also come to school to play, study, or interact with their friends and teachers. Teachers will remain an important part of the school. They will oversee the students' performance, create recorded lectures, and provide one-on-one support to students who have difficulties in academics. If a student is struggling and falling behind in the curriculum, the teacher will make sure that their difficulties are addressed.

The goal will be to help the student complete the syllabus on time without rushing through or skipping topics. The breaks between classes, lunchtime recess periods, PE periods and other extracurricular activities will be same as in the existing traditional school models.

What we will build in Round 2

Round 2 focus

A small but complete adaptive-learning loop for one Classes 6–8 science unit on electricity and circuits. The prototype will use four connected concepts, two fictional student accounts and one teacher account.

- Student login and account isolation
- Lesson/reading interface and doubt-solving tutor
- Predefined first micro-test
- Typed or handwritten written answers
- AI misconception diagnosis
- Personalized support and reassessment
- End-of-day personalized homework
- Teacher-approved graphical/video resources for doubt clearing

The common end-of-term exam remains part of the school model, but it is designed and managed by teachers or the school. It is not generated by SAFAL.

1. Learning workflow

The prototype follows one clear learning loop. It is deliberately narrow so that the complete flow can be demonstrated live rather than presenting many unfinished features.

Stage	What happens
1. Lesson	Students open the recorded-lecture interface and reading material. The demo may use a sample or placeholder video, but the complete lesson UI is shown.
2. Doubts	SAFAL asks whether the student has doubts. If yes, the tutor answers using teacher-approved material and can open a relevant diagram, graphic or preloaded YouTube video when a visual explanation would help. If no, the student moves to the micro-test.
3. First micro-test	The first assessment is predefined and common to students. It uses written/open-ended questions rather than MCQs.
4. Analysis	SAFAL marks each answer right or wrong and analyses the reasoning even when the final answer is correct. Handwritten work can be uploaded as an image.
5. Support	SAFAL explains where the student went wrong, identifies a likely misconception and gives targeted teaching or practice.
6. Reassessment	A new personalised reassessment is generated from the diagnosed gap.
7. Second support cycle	If the student still struggles, SAFAL tries one more targeted support and reassessment cycle.
8. Outcome	If the student succeeds, SAFAL records the result and the teacher decides when they move to the next concept. If they still struggle after the second support cycle, SAFAL flags them for teacher attention.

Assessment rules for the prototype

- No numerical score is shown. Each response is marked right or wrong.
- Correct answers are still checked for faulty reasoning or lucky conclusions.
- The first micro-test is predefined; later reassessments are personalised and AI-generated.
- A second fresh check may exist in the full SAFAL model, but it is not part of the Round 2 prototype.

2. Diagnosing learning gaps

The main AI feature in SAFAL is not simply checking whether an answer is correct. It tries to understand why the student answered that way. This is what allows two students who make different mistakes on the same topic to receive different support.

What SAFAL analyses

Input	Use
Question and expected idea	Defines what understanding is being checked.
Student answer	Shows the final response.
Student reasoning or working	Provides evidence of how the student reached the answer.

Handwritten submission	A vision-capable AI reads the uploaded work and interprets the reasoning.
Relevant earlier responses	Helps avoid treating one isolated mistake as the whole picture.
Approved textbook/reference material	Keeps the diagnosis and teaching within the teacher-defined curriculum.

Diagnosis output

The AI returns a small, structured result that the application can store and display. We only need fields that are useful to the learning flow:

Example diagnosis

Concept: Complete circuits
Result: Wrong
Likely misconception: Student does not yet understand that a complete conducting path is required.
Evidence: The student expects the bulb to light without a return path.
Next action: Ask a short follow-up, then explain open vs. closed circuits and give targeted practice.

If the reasoning is unclear, SAFAL asks a short follow-up question instead of pretending to know the cause. The teacher can override the diagnosis or support at any time.

Handwritten work

For Round 2, handwritten submission is kept simple: the student uploads an image of their work, the AI interprets it, and the interpreted answer enters the same diagnosis flow as a typed response. We are not building a separate handwriting-recognition system.

3. Personalised support and homework

Support after a mistake

After the micro-test, SAFAL first tells the student which responses were right or wrong. It then uses the diagnosed gap to choose a suitable explanation, guiding question, example or practice task. The aim is to repair the specific misunderstanding rather than repeat the whole lesson.

All teaching and generated questions are restricted to material approved by the teacher. This can include textbooks, reference material, diagrams, graphics and selected YouTube videos. During doubt clearing, SAFAL can choose the most useful approved visual or video for the student's question. If the school permits it and the approved library is insufficient, SAFAL may also retrieve supplementary visual material from approved online sources.

Personalised reassessment

The first micro-test is fixed. Reassessments are generated for the individual student so they can test the same idea in a fresh way. If the first support cycle does not work, SAFAL tries once more. Continued difficulty is then flagged for the classroom teacher.

End-of-day homework

At the end of the school day, SAFAL generates a personalised homework assignment using the weaknesses it observed across that day. Teacher approval is not required for every assignment in the prototype, because that would slow the system down too much.

Students can open and submit homework from home through the same SAFAL web portal and credentials. Submitted homework is analysed using the same answer-and-reasoning process. Its results can affect later explanations, practice and reassessments.

Teacher role

The AI handles routine diagnosis, support, reassessment and homework generation. The teacher remains the final authority. There is no complicated override procedure: if the teacher believes SAFAL is wrong or wants a different approach, they can intervene directly.

4. Teacher view and student access

Teacher dashboard

The teacher dashboard is intentionally basic. It shows the two students, the concept they are working on and whether SAFAL has flagged them for attention. It does not include term pacing, rankings or complex analytics.

Student	Current concept	Status
Student A	Complete circuits	Working normally
Student B	Complete circuits	Teacher attention needed

Opening a student shows the evidence already collected: their written or handwritten answers, the likely misconception, support SAFAL gave, and the reassessment result. This lets the teacher understand the problem without reading through the student's entire session.

Accounts and access

The prototype uses basic authentication so one student cannot open another student's account or learning data. Students access their own records; the teacher can view the fictional class. API keys and privileged database access stay on the server.

The classroom version is designed for school tablets, while the same web portal can be opened from home for homework.

5. Software design

We will build SAFAL as one web application with separate student and teacher views. The design is kept simple so the team can focus on the learning loop rather than infrastructure.

Part	Proposed technology	Purpose
Web application	Next.js + TypeScript	Student and teacher screens, server routes and learning-flow logic.
Database and login	Supabase PostgreSQL + Auth	Accounts, student responses, learning state, homework and teacher view.
AI layer	Server-side AI API	Doubt solving, handwriting interpretation, misconception diagnosis, personalised support, reassessment and homework generation.
Approved content	Teacher-selected textbooks, references and visual resources	Approved material the tutor may use for teaching, including diagrams, graphics and selected YouTube videos. Optional online retrieval can be limited to approved sources.

Main server actions

Action	What the server does
Submit answer	Checks the logged-in student, saves the response, marks it right/wrong and sends relevant evidence for AI analysis.
Ask tutor	Sends the student's doubt with the approved material for the active concept and can select a relevant approved diagram, graphic or video for visual explanation.
Upload handwritten work	Stores the image, asks the AI to interpret it and passes the interpreted work into the normal analysis flow.
Generate reassessment/homework	Uses the diagnosed gaps and approved sources to create fresh written questions.
Open teacher view	Returns the two students' current state and any teacher-attention flag.

Failure handling

If the AI fails to return a usable response, the student's submitted work is preserved and the case can be referred to the teacher.

6. Round 2 prototype and demo

Science unit

The demo uses four connected electricity-and-circuit concepts for Classes 6–8. We will keep a small, reviewed set of starting questions and teacher-approved learning material. There is no fixed question-bank size.

Concept	Learning objective
C1. Sources and terminals	Identify a cell's terminals and basic parts of a simple circuit.
C2. Conducting materials	Distinguish common conductors and insulators in the example circuit.
C3. Complete circuits	Explain why a complete conducting path and closed switch matter.
C4. Finding faults	Locate a break or unsuitable material in a simple circuit.

Screens to demonstrate

- Student login and current-task screen.
- Lesson page with video area, reading material and the "Do you have any doubts?" step.
- Live AI tutor for doubts, with a graphical explanation view that can show teacher-approved diagrams, graphics or animated YouTube videos when a visual explanation would help.
- Written micro-test with typed answer or handwritten-image upload.
- Feedback, misconception diagnosis, personalised support and reassessment.
- End-of-day personalised homework that can be opened/submitted from home.
- Teacher dashboard with student detail view and attention flag.

Demo journey

The two fictional accounts can be reset and rerun with different responses. In one run, Student A can misunderstand the need for a complete circuit while Student B confuses a conductor and an insulator. SAFAL should diagnose the different reasoning, give different support and generate different reassessments. A student who continues to struggle after two support cycles is flagged on the teacher's dashboard.

7. Building with Cline

Cline is the core development tool for Round 2. We will give it the project requirements, ask it to plan small parts of the system, let it create or edit the code, run the application, and fix issues as we test the flow. The AI used inside SAFAL is a separate model API called by the finished application.

Milestone	What we ask Cline to build
1. Foundation	Next.js project, Supabase connection, login, two student accounts, one teacher account and four-concept seed content.
2. Student flow	Lesson UI, doubt step, graphical/video explanation view, predefined micro-test, written answers and saved state.
3. AI loop	Doubt-resource selection, handwriting interpretation, misconception diagnosis, targeted support and personalised reassessment.
4. Homework	End-of-day personalised homework and home submission.
5. Teacher view	Simple dashboard, student evidence views and teacher-attention flag.
6. Demo pass	Connect the full flow, reset/reseed the two demo accounts and fix issues found during rehearsal.

Project rules for Cline

- Use only fictional student data in the demo.
- Keep API keys and privileged database access on the server.
- Keep each student's data isolated by login.
- Restrict tutoring, generated questions and visual resources to teacher-approved sources; any optional web retrieval must follow the same restriction.
- Teachers must always be able to override SAFAL's recommendations.